

Benchmarking sequence, embedding and structure-based methods for eukaryotic virus protein annotation

Lander De Coninck^{1*}, Uri Neri², Susie Grigson², Dimitris Karapliafis³, Simon Roux²

¹KU Leuven, Microbiology, Immunology and Transplantation, Clinical and Epidemiological Virology, Herestraat 49, box 1040, Leuven, 3000, Belgium.

²Lawrence Berkeley National Laboratory, DOE Joint Genome Institute, Cyclotron Road 1, Berkeley, CA 94720, USA.

³Wageningen University, Bioinformatics Group, Droevendaalsesteeg 1, Wageningen, 6708PB, The Netherlands.

*Corresponding author(s). E-mail(s): lander.deconinck@kuleuven.be;

Contributing authors: uner@lbl.gov; sgrigson@lbl.gov; dimitris.karapliafis@wur.nl; sroux@lbl.gov;

Abstract

The advent of protein language models and structural prediction promises new avenues for protein annotation through structure homology. Earlier work (eg. Phold) has shown protein structural homology annotation works well for prokaryotic viruses. To leverage these new techniques for protein annotation of eukaryotic viruses, we benchmarked four recent embedding- and structure-based methods in comparison with the gold standard of sequence-based annotation. We selected 10% of the entries in ICTV's VMR v40.2 that did not have bacteria or archaea as designated host (n=1,054), and subsequently retrieved all associated proteins from NCBI (n=11,360). Boltz-2 structure prediction of these proteins produced generally high-quality models (median pLDDT 71.3). Using the structures and protein sequences of the entries in the Big Fantastic Virus Database, we queried the VMR proteins with BLASTp, Diamond, and MMseqs2 (sequence); TEA and ProstT5-3Di (embedding); and Foldseek and Reseek (structure). Sequence homology-based methods showed a high overlap in identical hits, whereas the other methods had low intra- and inter-method agreement in hits between themselves and the sequence-based methods. Additionally, all methods improved the number of informative categories (ie. not "hypothetical protein") for the VMR proteins, with embedding and structure methods producing the largest increases. However, when evaluated against proteins with informative GenBank annotations as ground truth (n=3,983), embedding- and structure-based methods agreed less often with the "ground truth" than sequence-based methods (maximum 63.6% and 72.2% vs 78.4%, respectively). Overall, sequence homology remains the most accurate approach for protein annotation of eukaryotic viruses at present, although for a minority of "hypothetical proteins" in mostly large dsDNA viruses structure homology could help resolve functional annotation.

Introduction

The current increase in metagenomic sequencing experiments is providing researchers with a steady stream of new viral genomes. A large problem that remains for computational biologists is the annotation of these genomes with functionally relevant labels for the predicted proteins. As annotating all these genomes through experimental evidence is an unfeasible task, researchers transfer annotations of annotated genomes to these novel genomes. The established gold standard for performing this task has been the use of the BLAST suite or comparable sequence alignment tools, such as Diamond and MMSeqs2¹⁻³. However, this approach reaches its limits when newly discovered sequences diverge too much from experimentally validated annotated genomes, a problem that is particularly acute in virology, where mutation rates are high and sequence diversity is vast⁴.

Protein language models (pLMs) have emerged as a powerful paradigm for capturing biologically meaningful representations from amino acid sequences alone, trained on known protein sequences using self-supervised objectives analogous to those used in natural language processing^{5,6}. By learning the statistical regularities of protein sequences across millions of examples, pLMs implicitly encode evolutionary, structural, and functional information in dense embedding vectors. These representations have proven useful for a wide range of downstream tasks, including secondary structure prediction, contact map prediction, solubility prediction, and functional site annotation⁵⁻⁷. Models such as ESM-2⁷ and ProtTrans⁶ have become widely used as general-purpose sequence encoders in the field.

A complementary and increasingly important line of research leverages protein three-dimensional structure, which is generally more conserved than primary sequence over long evolutionary timescales, because the topology of a folded protein is directly tied to its biochemical function⁸. The advent of highly accurate structure prediction tools such as AlphaFold2⁹ and, more recently, Boltz-2¹⁰, has dramatically lowered the barrier to obtaining structural models. Structural comparison tools, most notably Foldseek¹¹, have made it feasible to search large structural databases at scale by encoding local backbone geometry into a compact structural alphabet (3Di), enabling alignment-based searches analogous to BLAST but operating on structure rather than sequence. More recently, Reseek¹² was developed to further improve sensitivity to remote structural homologs.

The convergence of pLMs and structural biology has opened a third avenue: using language models to predict structural information without the computational cost of three-dimensional structure prediction. ProstT5 is a notable example, fine-tuning a protein language model to translate amino acid sequences into 3Di token sequences, effectively producing a structural representation with a three order magnitude speedup compared to structure prediction¹³. A related approach, The Embedded Alphabet (TEA), uses ESM-2 embeddings as input to generate an alternative structural alphabet that the authors argue better represents helices and loops

74 compared to 3Di¹⁴. These compressed one-dimensional structural representations can then be
75 used as inputs to standard alignment tools, bridging the gap between pure sequence methods
76 and full structure comparison pipelines.

77 To address the annotation bottleneck for phage genomes specifically, Phold¹⁵ cleverly com-
78 bines structural information with a large database of high-quality phage protein predictions.
79 Phold either accepts user-provided structure predictions or uses ProstT5 to convert amino
80 acid sequences into 3Di sequences, which are then aligned against annotated references using
81 Foldseek. Benchmarking showed that Phold improved the annotation of a select set of bac-
82 teriophage proteins from approximately 35% to approximately 50% with both ProstT5 and
83 protein structures¹⁵. A similar approach, Baktfold, was subsequently developed by the same
84 group to increase annotation rates in bacterial genomes¹⁶.

85 In addition to bacteriophages, also less studied eukaryotic viruses (e.g. insect viruses, nematode
86 viruses and plant viruses) might benefit from such an approach to bring a more functional
87 understanding to metagenomic discoveries of these viruses. For example, the *Chrysoviridae*, a
88 family infecting fungi, plants, and likely insects, whose four dsRNA genome segments encode
89 four proteins. Two of these proteins — an RNA-dependent RNA polymerase and a capsid protein
90 — have established functions, while the remaining two are of entirely unknown function¹⁷.

91 Therefore, we benchmarked seven approaches for functional annotation: BLASTp, Diamond and
92 MMseqs2 as gold standard sequence alignment methods; ProstT5 3Di conversion with Foldseek
93 alignment, and ESM-2 to The Embedded Alphabet (TEA) conversion with MMseqs2 alignment
94 as pLM-derived structural alphabet approaches; and finally, full three-dimensional structure
95 prediction of viral proteins with Boltz-2¹⁰ followed by structural alignment using either Foldseek¹¹
96 or Reseek¹². To evaluate these methods on a representative benchmark, we downloaded
97 11,360 proteins from 1,000 randomly selected, ICTV-ratified eukaryotic virus species, whose
98 formal taxonomic recognition implies a better degree of annotation than randomly selected
99 viral sequences from GenBank, while also being less comprehensively characterized than the
100 curated entries in RefSeq. For all methods, we used the Big Fantastic Virus Database (BFVD),
101 which contains 351,242 high-quality predicted viral protein structures, as a unified reference
102 database¹⁸, ensuring that each annotation method was evaluated against an identical set of
103 viral proteins and that observed differences in performance could be attributed to the annotation
104 strategy rather than to differences in the underlying reference.

105 Results

106 Boltz-2 generates high-quality viral protein structures without parameter 107 fine-tuning

108 Our test set of protein sequences consisted of 1,054 different viral entries taken at random from
109 ICTV's VMR data (n=1,000 distinct species). These test dataset viruses covered all eukaryotic

110 viral Kingdoms, with the majority of proteins originating from *Orthornavirae* (RNA viruses) and
 111 *Bamfordvirae* (dsDNA viruses) (see Figure 1A), despite the latter only representing 43 viruses
 112 (see Supplementary Figure 1).

113 Structure prediction of all 11,467 proteins from our test set with Boltz-2 resulted in high-quality
 114 predicted structures, with a median pLDDT score of 71.3 and 52.4% of all predicted structures
 115 having a pLDDT score above 70 (see Figure 1B). This shows that the structure prediction with
 116 default settings of Boltz-2 results in reliable structure predictions without the need for extensive
 117 fine-tuning of parameters, although extra viral sequences were used for the multiple sequence
 118 alignment (MSA) generation (see methods).

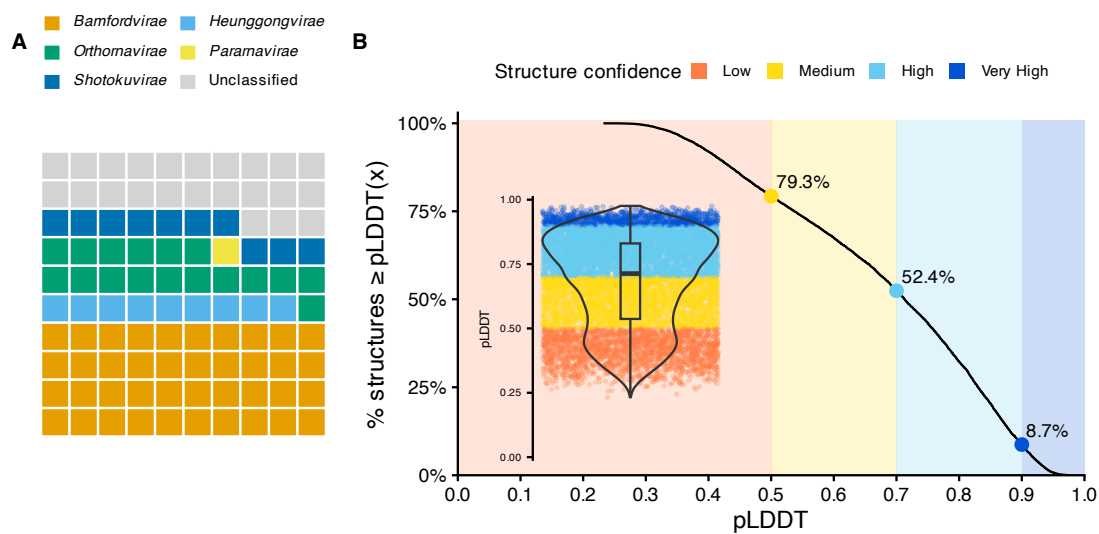


Figure 1: A) Test protein distribution from the different viral Kingdoms. Each tile roughly corresponds to 110 proteins. B) A cumulative distribution function plot showing the percentage of predicted structures from the test set with a pLDDT score above the pLDDT score on the x-axis. The inset shows a violin- and boxplot with all pLDDT scores from the predicted structures (n=11,467).

119 Sequence homology methods demonstrate superior sensitivity and repro- 120 ducibility over structure-informed approaches in eukaryotic virus annota- 121 tion

122 We started by evaluating how many of the proteins in our test produced at least one match
 123 against the BFVD for each method (see Figure 2A). In addition, we also filtered the results
 124 of each method on an E-value lower than 1e-03 to see how this affected the number of hits
 125 for each test protein. This decreased the number of test proteins with a hit on average with
 126 25% for the structure methods. The methods relying on structure homology clearly had less

127 matches compared to the sequence homology-based methods. This suggests that sequence-
 128 based methods are still more sensitive in detecting homology for eukaryotic viruses than
 129 structure-based methods, despite the promising results for bacteriophages¹⁵. As the quality
 130 of our structure predictions was rather high (see Figure 1B), this lower sensitivity is likely not
 131 due to poor structure predictions. In contrast, TEA had, as only non-protein sequence-based
 132 method, a comparable number of test proteins with a match, almost reaching the total number
 133 of test proteins (n=11,360). However, analyzing the total number of hits at different E-value
 134 thresholds shows that the TEA search resulted in a very high number of matches across all
 135 E-value thresholds, signifying a high number of false positives (see Figure 2B). A similar pattern
 136 was observed for the other structure-based methods, which tended to find too many hits at
 137 high E-values. Reseek was less affected by this, due to a combination of already calculating
 138 E-values more accurately^{12,19} and our use of a modified Reseek version (see below). On the
 139 other hand, the three sequence-based methods had a similar number of test proteins with a
 140 hit (BLASTp=11,359; MMseqs2=11,255 and Diamond=11,224), but don't show this inflation of
 141 total hits at higher E-values (BLASTp had a slight increase towards larger E-values).

142 When comparing the BFVD matches yielded by all methods against each other (see Figure 2C-D
 143 and Supplementary Figure 3), we observed that the sequence homology-based methods had a
 144 high overlap in identical hits. This was true when comparing the best 25 hits for each test protein
 145 (only the comparison of diamond and mmseqs2 against blastp had a low overlap because of
 146 the higher amount of hits for blastp), the best 25 hits afterwards filtered on E-value, and only
 147 the top hits (see Figure 2C-D and Supplementary Figure 3). In contrast, the embedding and
 148 structure homology-based methods had low intra-method agreement in resulting hits (~25%
 149 between the embedding methods and ~40-75% for the structure methods on the best 25 hits).
 150 The percentage of overlapping hits did increase when only taking the top hits into account,
 151 but stayed well below the percentage of overlap between the sequence-based methods. It is
 152 worth noting that Reseek seemed to perform worse in this metric when looking at the filtered
 153 results (see Supplementary Figure 3B), however this is due to the large reduction in number
 154 of matches after filtering. This reduction is due to the more stringent E-value calculations in
 155 Reseek compared to Foldseek^{12,19}. Notably, while ProstT5 was the worst performer across
 156 the three datasets compared to the sequence methods, TEA performed better in the top hits
 157 comparison and was on par with the structure-based methods. Overall, the embedding and
 158 structure-based methods, that should be better to detect remote homologs, had low agreement
 159 on shared hits among each other.

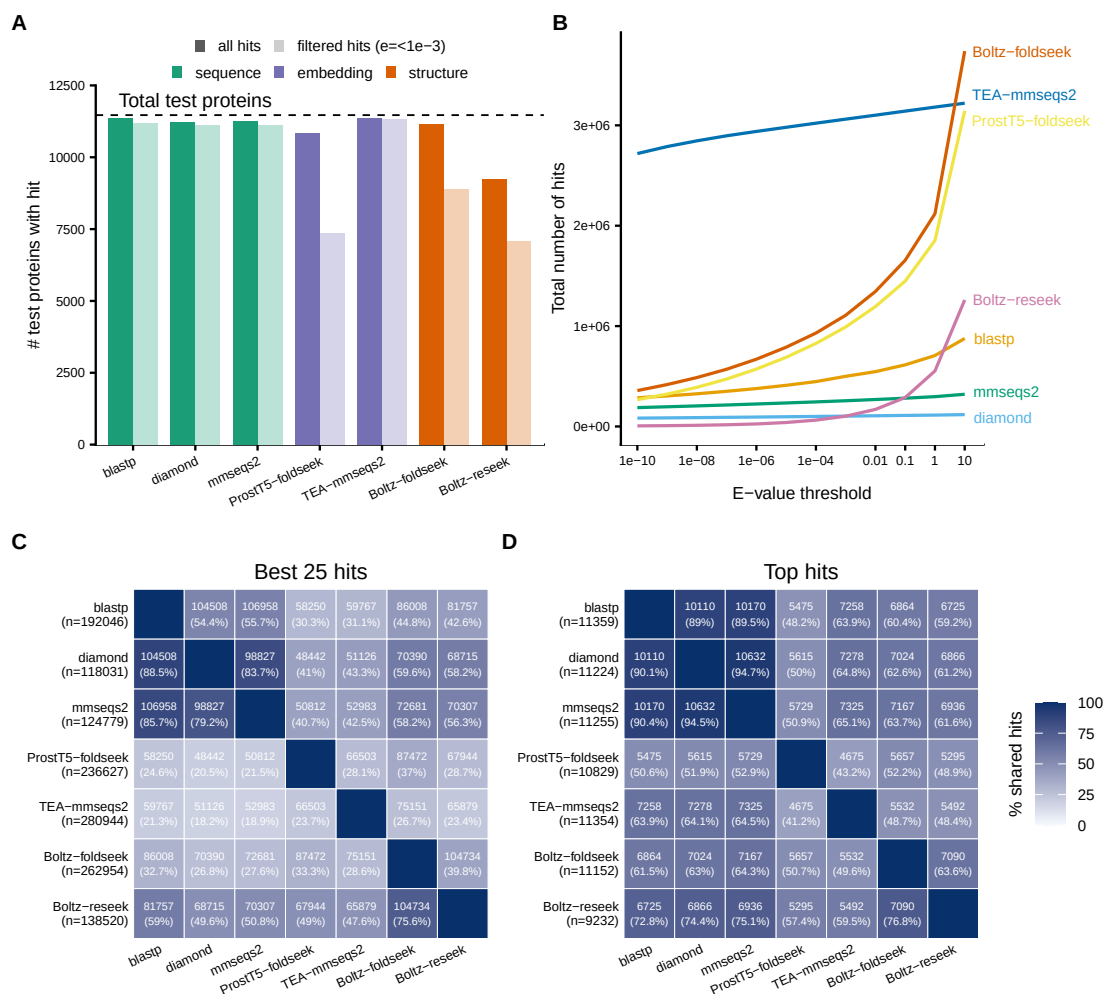


Figure 2: A) The number of proteins that got a hit against the BFVD shown per method (colored by category) before and after filtering the reported results on E-value lower than $1e-03$ (lighter color). B) The total number of hits across all test proteins in function of the E-value for each method. C-D) Pairwise comparisons of shared hits across methods for the same test proteins, shown for the best 25 hits (C), and the top hits regardless of E-value (D). Text inside the colored tiles indicate the number of shared hits for the method on the x-axis compared to the method on the y-axis, followed by the percentage of shared hits as a proportion of the total hits for the method on the y-axis.

Sequence-based methods retain better accuracy in functional category assignment while structural approaches slightly improve novel annotations

Using a weighted approach, we assigned functional categories to the test proteins for each method based on the hits with an E-value smaller than $1e-03$ (see methods). To evaluate the accuracy of the benchmarked methods, we considered the test proteins with an originally informative annotation ($n=3,983$) as the ground truth to compare our results to. All methods

improved the number of informative categories (see Figure 3A), with the sequence-based methods, together with TEA and Boltz-Foldseek, improving the number of informative assignments the most. However, when comparing the functional annotations to the “ground truth”, all embedding and structure-based methods underperformed compared to the sequence-based methods (maximum 63.6% and 72.2% vs 78.4% agreement with “ground truth”; see Figure 3B).

Although TEA made the largest improvements in assigning previous uninformative proteins to informative categories, it also showed the greatest inconsistencies between assigned informative categories that were already informative in the original test set (see Figure 3C). In addition, it also has 558 queries that were changed from an informative function to a non-informative one, which is almost double the amount compared to blastp which had the second highest informative-to-uninformative reassignments (see Figure 3C and Table 1). The same applies to the number of test proteins that originally had an informative category, but was assigned a different category with TEA (n=1,062). Furthermore, due to the likely high number of false positives in TEA hits (see Figure 2B), we concluded that the TEA uninformative-to-informative assignments lacked reliability.

Of note, Boltz structure prediction combined with Foldseek did assign more informative categories than the sequence-based methods to originally non-informative proteins (n=1,578 vs ~1,460; see Table 1). However, this method retrieved less test proteins with hits against the BFVD. Interestingly, 337 proteins were assigned an informative category with Boltz-foldseek that were uninformative with either of the sequence-based methods. Most of these proteins belonged to the *Bamfordvirae*, a kingdom including nucleocytoplasmic large dsDNA viruses (NCLDV) characterized by their massive genome sizes. Furthermore, also proteins from the *Heunggongvirae* kingdom were overrepresented compared to their presence in the total dataset (see Figure 1A and Figure 3D). The eukaryotic viruses in the *Heunggongvirae* consist of members of the *Herpesvirales*, which also have large dsDNA genomes. In addition, the quality of the predicted protein structure was an important factor in the assignment of an informative category by Boltz-foldseek as >90% of these proteins had at least a medium quality structure, while across the structure prediction of all proteins this was 79.3% (see Figure 1B and Figure 3E). For the high-quality structures, this percentage even almost doubled (15% vs 8.7%). Finally, we looked at the categories these proteins belonged to and found that most (~40%) of them belonged to proteins with a structural function (see Figure 3F; includes both Structural and pvp - phage virion proteins). Also, ~10% of the proteins were assigned an “Anti-viral evasion” function which was mostly driven by Ser/Thr-kinases and interferon antagonist assignments.

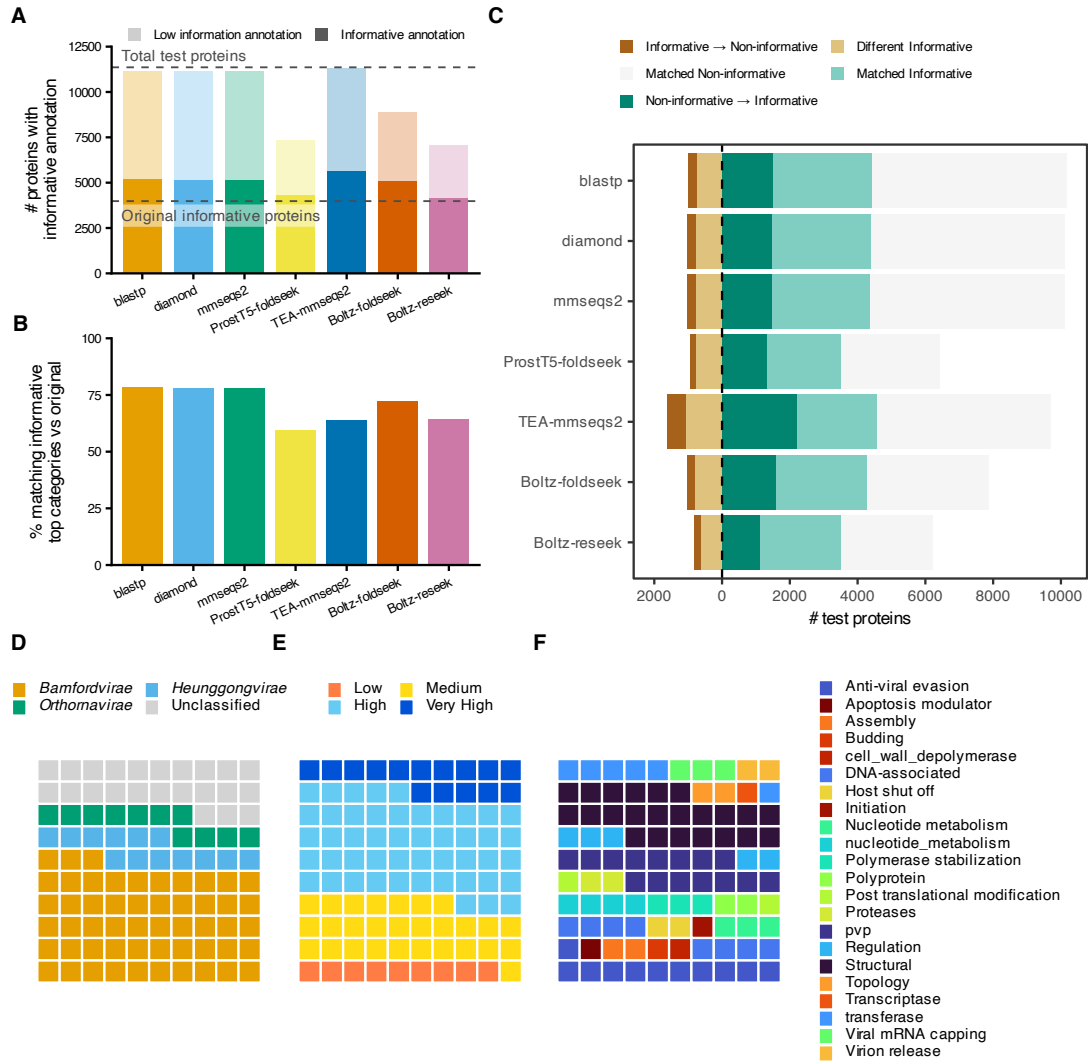


Figure 3: A) The number of proteins that received an informative functional annotation (i.e. not “Hypothetical Protein”, “unknown” or “Other Function”) after applying each method. Colored by method, light shades indicate non-informative proteins, dark shade indicates informative protein assignments. B) The percentage of queries that have a matching functional category for each method compared tot the informative original functional categories (regarded as the ground truth). C) Comparison of final assigned annotation categories compared to the original categories. Negative changes (ie. different informative categories or informative labels that changed to non-informative categories) are displayed to the left of the zero point, matches or category improvements (from non-informative to informative) are displayed on the right. D-F) Waffle plots showing the proportions of proteins that could be assigned an informative function with Boltz-foldseek and not with the sequence based methods, colored by D) taxonomy on Kingdom level, E) structure quality and F) main activity the assigned functional category takes part in (starts with uppercase; based on Table 2) or the main category of the Empathi hierarchical classification system (starts with lowercase). Tiles are ordered from bottom to top, and from left to right. Each tile corresponds to roughly three proteins.

Table 1: Functional annotation outcomes across the seven benchmarked methods.

method	Queries assigned	Informative match	Uninformative match	Informative mismatch	Uninformative to informative	Informative to uninformative
blastp	11177	2946	5740	757	1477	257
diamond	11119	2934	5710	765	1462	248
mmseqs2	11135	2930	5731	768	1452	254
ProstT5-foldseek	7366	2198	2902	789	1313	164
TEA-mmseqs2	11318	2358	5132	1062	2208	558
Boltz-foldseek	8889	2703	3592	807	1578	209
Boltz-reseek	7065	2408	2710	635	1116	196

Computational benchmarking

All search methods were benchmarked for runtime and memory usage using Hyperfine²⁰, and `nvidia-smi` for GPU memory consumption. As expected, the sequence-based methods were much faster and less memory intensive compared to the embedding and structure-based methods (see Figure 4 and Supplementary Figure 4). Both Diamond and MMseqs2 finished the search of the >11,000 proteins against the BFVD in under 10 seconds, followed by BLASTp with a runtime of almost two minutes, a ~28x time increase compared to MMseqs2 which was the fastest workflow. The embedding methods took 52 and 93 times longer than MMseqs2 for TEA and the ProstT5 workflow, respectively. Sequence-based methods were generally compatible with consumer-grade hardware. In contrast, the TEA workflow using the full precision ESM2 model required substantially greater GPU VRAM, necessitating specialized hardware, whereas ProstT5 operated within the memory constraints of consumer-grade GPUs. The structural workflows, by contrast, required over 20,000-fold more runtime than the sequence-based tools (see Supplementary Figure 4) and demanded substantial computational resources, exceeding 400 GiB of system memory (CPU) and approximately 80 GiB of GPU memory (see Figure 4). Furthermore, the Reseek structural comparison incurred an additional two-hour runtime, whereas the Foldseek-based structural comparison completed in approximately two minutes.

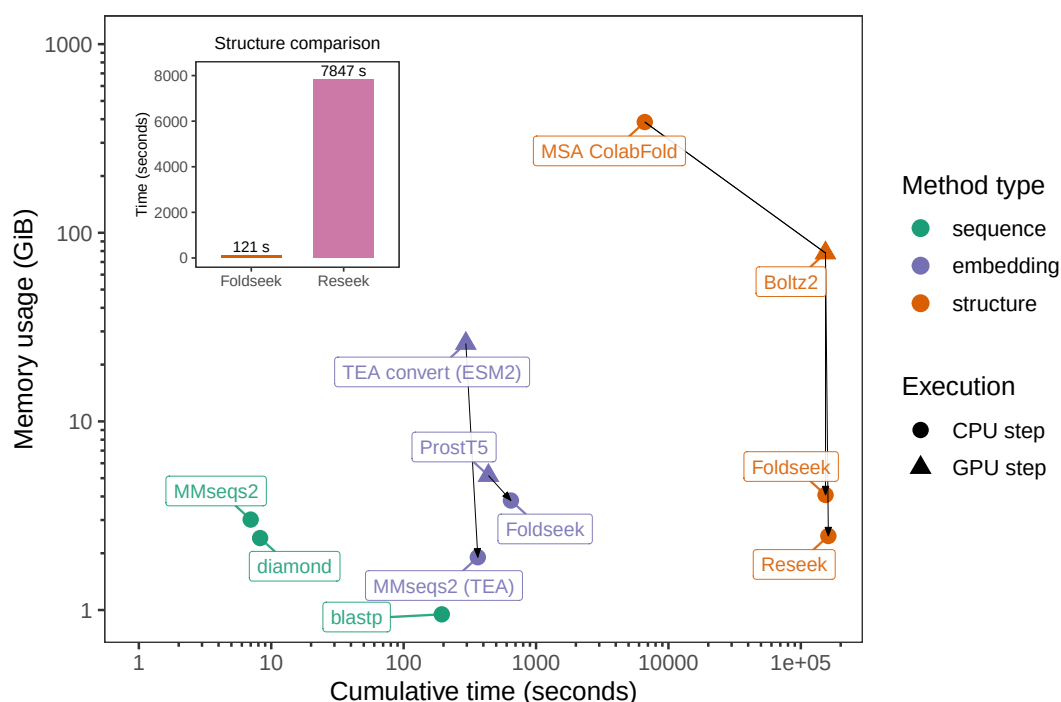


Figure 4: Log₁₀-transformed runtime in seconds of the different workflows plotted in function of the log₁₀-transformed memory usage (in GiB). For multistep workflows the indicated time is a cumulation of all steps (arrows indicate directionality of workflow steps). Workflows are colored by search method, CPU and GPU steps are indicated by circles and triangles respectively. The inset shows the difference in runtime between foldseek and reseek after MSA generation (ColabFold) and structure prediction (Boltz2).

Discussion

In this study, we benchmarked several established and novel methods for annotating eukaryotic viral proteins with a function. Our goal was to evaluate these tools using largely default settings, reflecting their usability in high-throughput annotation workflows by experienced researchers without specialized expertise in each method. In this regards, all tools performed well and were easy to run due to their extensive documentation. Most notably, structure prediction with Boltz-2 produced high-quality models without extensive parameter tuning. Aside from expanding the ColabFold MSA with additional viral sequences, all runs were performed using default settings, indicating that generating accurate structures is accessible even to researchers with limited experience in structure prediction.

Although new methods like pLMs and structure homology searches have shown great promise in detecting remote homologs^{11,12,14}, including bacteriophage protein searches¹⁵, our benchmark suggests that these advantages do not yet translate straightforwardly to eukaryotic virus proteins. Across our dataset, the sequence-based methods were consistently the most reproducible

(high overlap of identical hits) and the most accurate when evaluated against the original classifications, whereas embedding- and structure-based methods showed lower agreement with the ground truth and substantially lower concordance with one another.

A possible explanation for the low performance of the pLM approaches used here could be the lack of (eukaryotic virus) training data that was used to train these models due to an underrepresentation of viruses in large protein databases (e.g. UniRef90) and biosafety concerns^{21–23}. The use of a virus-specific pLMs, like LucaVirus²¹, could potentially mitigate this issue. Although the LucaVirus pLM is already available to generate the embeddings, their newly developed tool, LMalign, to align the generated embeddings is not. This did not readily allow us to compare the test protein embeddings with embeddings from the BFVD, our unified reference to ensure a fair comparison. On the other hand, it should be noted that recent work on mutational effect prediction with fine-tuned pLMs on viral sequences found that fine-tuning pLMs did not completely resolve the observed performance issues of pLMs on viral datasets²⁴.

In contrast, the structure-based comparison with Foldseek outperformed the best pLM approach by ~9% when benchmarked against the ground truth, yet remained less accurate than sequence-based methods. Despite this limitation, structural comparison enabled the functional annotation of 337 previously hypothetical proteins that were not classified based on sequence approaches. However, this gain is modest relative to the broader improvements in functional annotation reported for bacteriophages using structure homology¹⁵. The majority of these newly annotated proteins originate from large dsDNA viruses and were assigned structural functions, which likely reflects the overrepresentation of structurally annotated proteins in the BFVD (see Supplementary Figure 5), thereby facilitating the detection of homologs among hypothetical proteins. Notably, approximately 10% of the hypothetical proteins receiving informative annotations were assigned the “Anti-viral evasion” label. Although this category is well-represented in the BFVD, its frequency among newly annotated proteins exceeds what would be expected relative to their presence in the BFVD. This implies that structure-based approaches may be particularly valuable for uncovering anti-viral evasion functions in large dsDNA viruses, consistent with evidence of conserved evasion strategies shared across phylogenetically unrelated virus families such as *Poxviridae* and *Paramyxoviridae*²⁵.

A few limitations arise from our approach: First, polyproteins — large precursor proteins that are subsequently cleaved by proteases into multiple smaller, functional proteins — were treated as single entities throughout our analysis. While this does not accurately reflect their biological reality, it was an inherent constraint imposed by the BFVD and UniProt, neither of which consistently provides information on individual subproteins within a polyprotein. Secondly, our LLM-based classification is susceptible to ambiguity when viral proteins with a distinct function share the same name. A notable example is VP1 (viral protein 1), which designates a capsid protein in *Parvoviridae*²⁶ but corresponds to the RNA-dependent RNA polymerase in *Sedoreoviridae*²⁷. To mitigate this, we provided taxonomic information to the LLM to guide

classification; however, classification errors were not completely avoided. For instance, when the name was spelled out in full as “viral protein 1”, the LLM failed to correctly classify the *Parvoviridae* VP1, instead assigning it the “Hypothetical protein” label. Nevertheless, even with these shortcomings, the LLM-based approach outperforms classical pattern-based matching in handling such ambiguities. Finally, the structural homology search was constrained by the composition of the BFVD, which currently contains approximately 350,000 structures, a small fraction of all known viral proteins. Nevertheless, the BFVD represents the largest available repository of eukaryotic virus protein structures, and expanding its coverage would represent a major undertaking, requiring substantial computational resources and time.

Overall, our benchmark supports a conservative conclusion: for eukaryotic virus protein annotation, sequence homology remains the most accurate and reproducible primary strategy. Structure-based methods serve as valuable complementary tools; however, their application should be deliberate and targeted given the substantial computational resources they require. The embedding-based methods and pLMs evaluated in this study do not offer sufficient accuracy for reliable functional annotation of eukaryotic virus proteins. While fine-tuned models may improve performance, their accuracy for functional annotation must be rigorously and continuously validated rather than assumed.

Material and Methods

Dataset selection

Test set

To benchmark different methods for eukaryotic virus protein annotation, we selected 10% of the entries in ICTV’s Virus Metadata Resource (VMR) v40.2 that did not have bacteria or archaea as designated host ($n=1,054$; 1,000 distinct species). We subsequently extracted their GenBank accession numbers ($n=1,402$) and downloaded all associated proteins ($n=11,360$).

Protein sequences that were longer than 2,500 amino acids were split into multiple chunks of maximum 2,500 amino acids so that structure prediction with Boltz-2¹⁰ would not run out of GPU memory (see below). This resulted in a final test set of 11,467 protein sequences from 1,000 eukaryotic virus species.

Database

The BFVD¹⁸ was used as the reference database for both sequence and structure homology-based annotation. The BFVD is a comprehensive database of high-quality predicted protein structures from the viral sequence representatives of the UniRef30 clusters. UniRef30 is a UniProt database that clusters protein sequences at 30% sequence identity and selects a representative sequence for each cluster, these representatives thus have a UniProt accession.

303 Since a substantial amount of BFVD's included proteins are no longer available in UniProtKB as
304 they are not part of reference proteomes, we converted their UniProt accessions to UniParc IDs.
305 The BFVD database was further converted into the required formats the search tools used in
306 this benchmarking effort (see sequence and structure homology-based annotation).

307 **Functional classifier**

308 Functional categories for each BFVD entry were derived from protein names associated with
309 the corresponding UniParc IDs (could be from UniProtKB/TrEMBL, RefSeq and/or GenBank) for
310 eukaryotic viruses or by classifying them into PHROG-like²⁸ categories with Empathi²⁹ in the
311 case of bacteriophage proteins. The distinction between eukaryotic virus and bacteriophage
312 proteins was made based on the TaxID information from the UniParc protein ID.

313 Functional categories for eukaryotic virus proteins were designed based on previously estab-
314 lished enzymes by ViralZone (see <https://viralzone.expasy.org/10417>) and adding structural
315 proteins together with extra enzymatic categories. These extra categories were acquired by
316 prompting Perplexity AI in deep research mode with:

317 "Can you give me an exhaustive list of common protein functions in eukaryotic viruses? Use [https://vi-](https://viralzone.expasy.org)
318 [ralzone.expasy.org](https://viralzone.expasy.org) and <https://ictv.global/report/genome> as sources. Give me a machine readable file
319 with all functional categories."

320 The resulting functional categories were further manually curated to ensure that they were
321 mutually exclusive. The final list of functional categories is provided in Table 2. A custom python
322 script was used to assign the established functional categories to the eukaryotic virus BFVD
323 proteins using Google's Gemma 4 31 billion parameter, instruction-trained LLM model locally
324 (<https://huggingface.co/google/gemma-4-31B-it>). The LLM had access to the collection of
325 protein names and the viral taxonomy of each entry, and was asked to provide a category
326 classification based on this information. Assigned functional categories were checked against
327 the originally established functional categories and removed if they did not correspond.

328 The same functional classifier script was also used on the protein names from our test set
329 to assign functional categories to the eukaryotic virus proteins downloaded from the ICTV
330 exemplar species.

331 An overview of the LLM pipeline is given in Supplementary Figure 6.

332 **Sequence homology-based annotation**

333 For sequence homology annotation, we used three state-of-the-art tools widely used in func-
334 tional annotation pipelines and sequence similarity searches. As gold standard, BLASTp
335 v2.16.0+¹ was used to align the test set amino acid sequences against the amino acid sequences
336 of the BFVD entries. In addition, we used `diamond blastp` (Diamond v2.1.16)³⁰ in sensitive

mode and `mmseqs easy-search` (MMseqs2 v18.8cc5c)³ with default settings; except for an E-value cutoff of 10 instead of 1e-03 to allow comparison to the other methods.

Protein embedding-based annotation

ProstT5

ProstT5¹³ is a pLM, based on ProtT5-XL-U50⁶, finetuned for translating amino acid sequences into structure representations with 3Di sequences. We used ProstT5 within `foldseek` v10.941cd33¹¹ using default settings of `foldseek createdb` to generate 3Di sequences for our test set and search them against the BFVD with `foldseek easy-search`.

TEA

In contrast to ProstT5, TEA¹⁴ uses the ESM2 pLM⁷ to generate embeddings for protein sequences and converts these embeddings into TEA sequences using this novel structural alphabet. Despite TEA and 3Di both consisting of 20 characters, TEA is different from 3Di in terms of the numbers and distribution these TEA characters map to secondary structural elements, promising a better representation of different structural elements across multiple alphabet characters.

We converted both our test set and the BFVD into TEA sequences using the `tea_convert` pipeline with default settings, and subsequently performed sequence similarity searches using `mmseqs easy-search`³ with a custom substitution matrix (matcha substitution matrix provided with TEA) and gap scoring settings (`--comp-bias-corr 0 --mask 0 --gap-open 18 --gap-extend 3`) to search the test set against the BFVD.

Structure homology-based annotation

Structure prediction

For the structure prediction of our test set proteins, we relied on Boltz v2.2.1¹⁰ (default settings) with MSAs generated by a local, adapted ColabFold installation³¹ (see <https://github.com/bouras13/colabfoldv>). To increase the depth and diversity of the MSAs, we used the default ColabFold sequence databases, supplemented with extra *Orthornavirae* sequences³² and viral sequences used in the structure prediction for the Phold database¹⁵.

Foldseek

Foldseek¹¹ converts predicted protein structures into 3Di sequences and subsequently performs fast structure similarity searches using these 3Di sequences. Here, we used `foldseek easy-search` (Foldseek v10.941cd33) with default settings to search the predicted structures of our test set against the BFVD.

369 **Reseek**

370 Reseek¹² aligns structures in a similar way to Foldseek, but uses a different structural alphabet
371 consisting of $\sim 10^{11}$ states, promising better sensitivity for remote homologs.

372 We used an adapted version of Reseek v2.7 in sensitive mode that switches the loading of the
373 queries and database to get more accurate E-value calculations (see [this discussion](#)).

374 **Functional category assignment**

375 To functionally annotate our test proteins for each method, weights in form of the negative log
376 E-value (or a weight of 1000 in case of an E-value of 0) were given to each hit. These weights
377 were summed per functional category (assigned to each entry in the BFVD; see above) and the
378 category with the largest weight was chosen as the functional category for each test protein.
379 Informative categories were prioritized over uninformative categories (ie. “Hypothetical Protein”,
380 “unknown”, “Other function”) when the weight of the informative category was at least 50% the
381 weight of the uninformative categories.

382 **Computational benchmarking**

383 All steps in our analysis were benchmarked for both time and memory usage using hyperfine
384 v1.20.0²⁰. GPU memory consumption was measured by sampling usage every 200 ms with
385 nvidia-smi v580.82.07, recording the peak value observed after each step.

386 All CPU analysis steps were performed on an HPC system using one IceLake node with 72 cores
387 (2 Intel Xeon Platinum 8360Y CPUs@2.4 GHz; 256 GiB RAM), except for the MSA generation
388 step which was performed on an IceLake big memory node with 72 cores (2 Intel Xeon Platinum
389 8360Y CPUs@2.4 GHz; 2048 GiB RAM). The GPU analyses were performed on one NVIDIA
390 H100 SXM5 GPU (80 GiB HBM3), mounted on a node equipped with 2 AMD EPYC 9334 CPUs
391 (768 GiB RAM).

392 **Code availability**

393 Code to reproduce this analysis is available on [GitHub](#).

394 **Data availability**

395 [upload to zenodo](#)

396 **Acknowledgements**

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398 V451025N). The resources and services used in this work were provided by the VSC (Flemish
399 Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish
400 Government.

401 Declaration of LLM usage

402 During the preparation of this manuscript, the authors made use of large language model (LLM)-
403 based artificial intelligence tools to assist with literature searches, restructuring and improving
404 sentences, and generating code for data analysis. All AI-generated content and code was
405 critically reviewed, verified, and edited by the authors, and full responsibility for the accuracy
406 and integrity of the work rests with the authors alone.

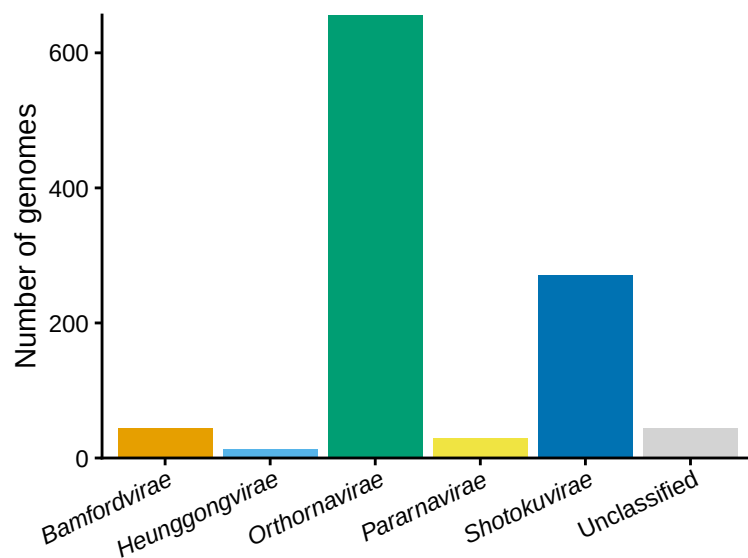
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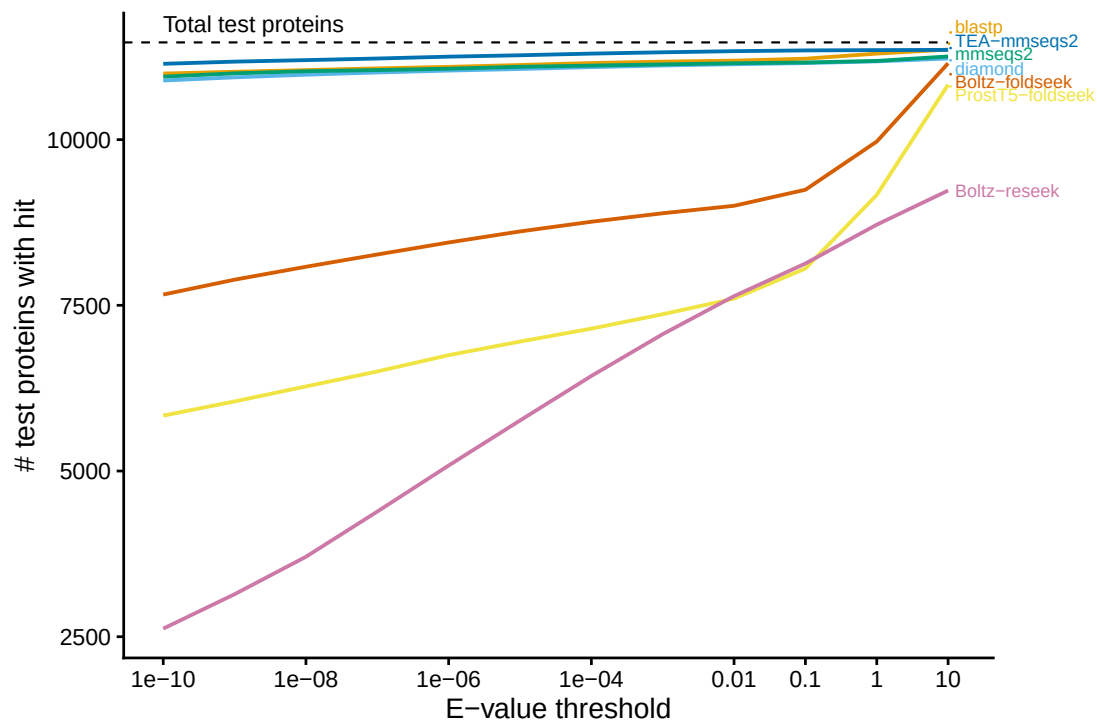
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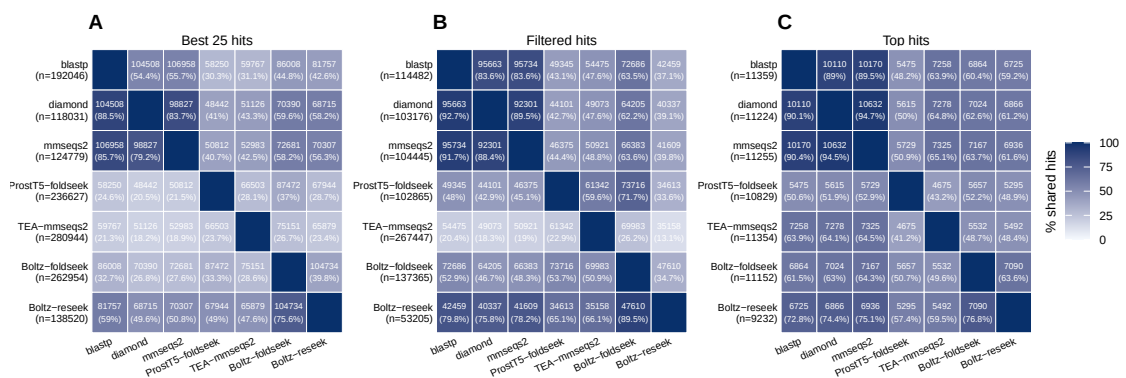
Supplementary Figures



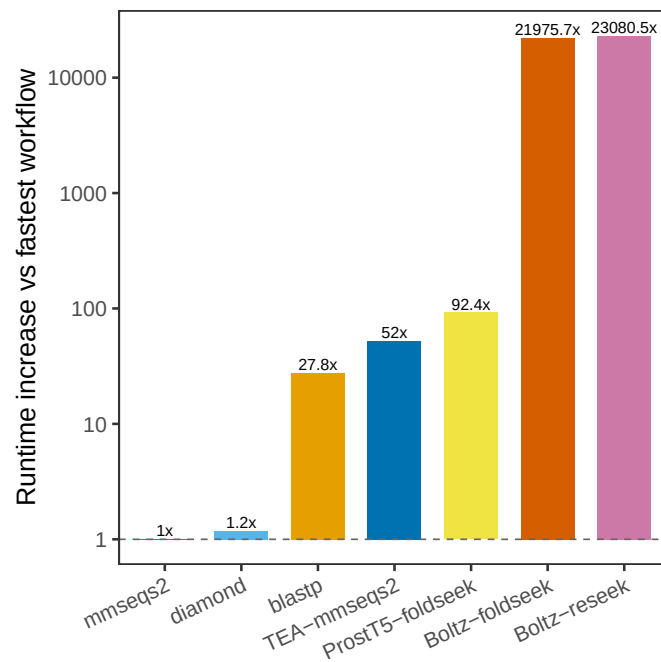
Supplementary Figure 1: The distribution of the number of genomes from the different viral Kingdoms in the test set.



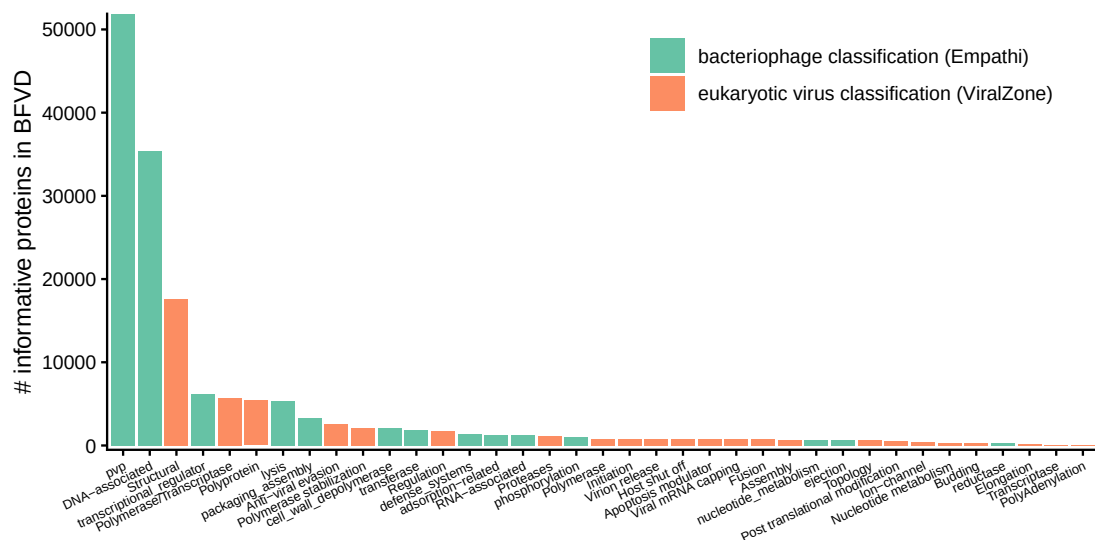
Supplementary Figure 2: The number of test proteins with at least 1 hit for each method across different E-value thresholds.



Supplementary Figure 3: Pairwise comparisons of shared hits between search methods for (A) the best 25 hits, (B) the best 25 hits after filtering on E-value ($E < 1e-03$), and (C) the top hits for each query.



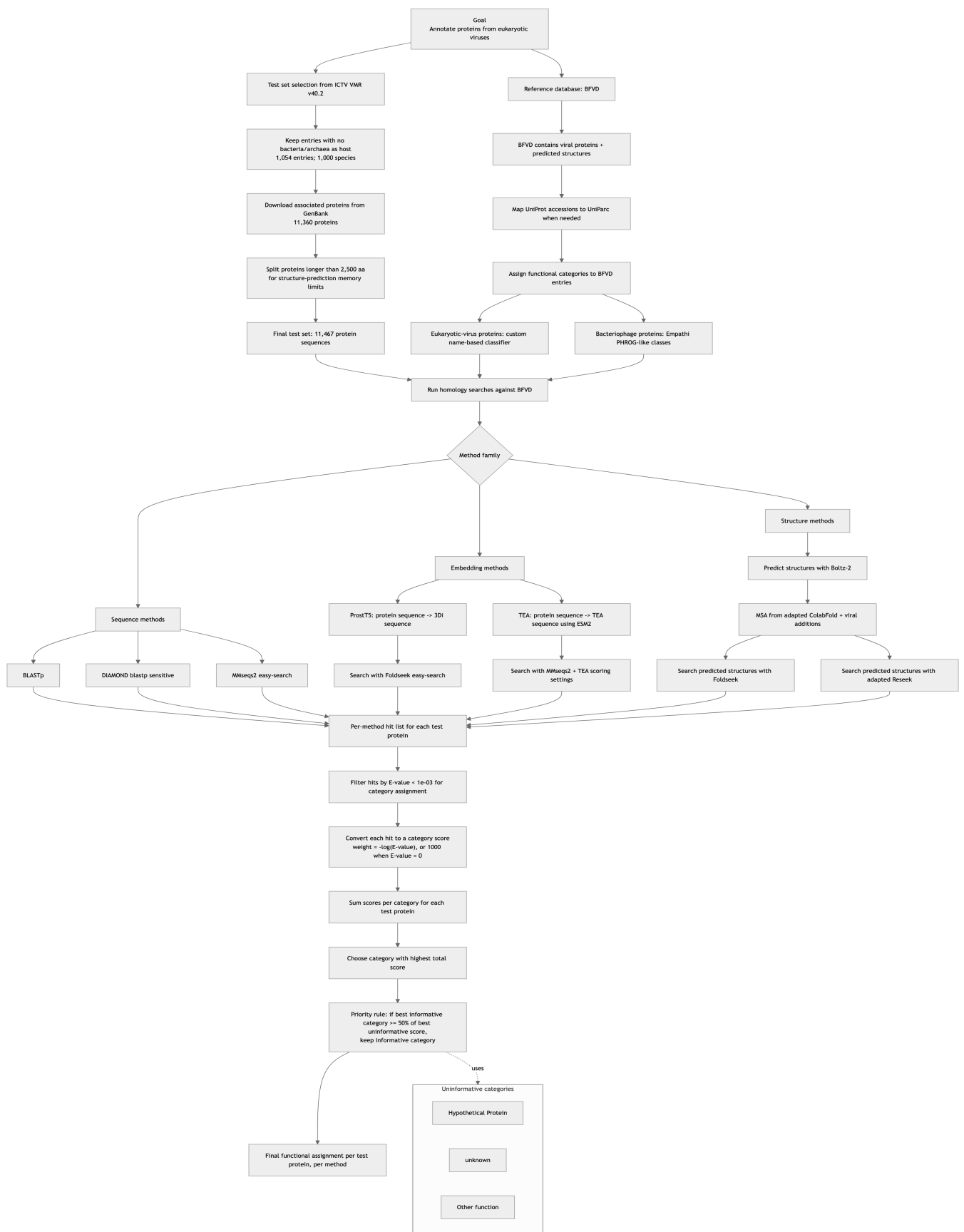
Supplementary Figure 4: Workflow runtime increase compared to the fastest workflow (MMseqs2).



Supplementary Figure 5: The number of informative proteins in BFVD by their main function, colored by eukaryotic or bacteriophage protein.



Supplementary Figure 6: Flowchart of the LLM classification pipeline for eukaryotic viruses based on the protein names in databases like GenBank, UniProt, etc.



Supplementary Figure 7: Flowchart of the annotation pipeline used in this study. Eukaryotic virus proteins were selected from ICTV VMR v40.2, filtered to exclude bacteria- and archaea-hosted entries, and expanded into a final test set after retrieving all associated protein sequences and splitting very long proteins for structure prediction. The BFVD reference database was prepared separately by assigning functional categories to its entries using a custom name-based classifier for eukaryotic virus proteins and PHROG-like categories for bacteriophage proteins by Empathi. The test proteins were then compared against BFVD using three sequence methods (BLASTp, DIAMOND, MMseqs2), two embedding-based methods (ProstT5-3Di and TEA), and two structure-based methods (Foldseek and Reseek after Boltz-2 structure prediction). Hits were filtered by E-value, converted into weighted category scores, summed per functional category, and the highest-scoring category was chosen for each test protein, with a rule that preserves informative categories when they are sufficiently supported over uninformative ones.

Table 2: Final eukaryotic virus functional categories used for annotation, including their activity and process they are involved in.

Process	Activity	Enzyme/Protein
Entry	Fusion	Fusion glycoprotein
Exit	Budding	Phospholipase
Exit	Virion release	Neuraminidase
Exit	Ion-channel	Viroporin
Exit	Budding	Virion release facilitator
Exit	Virion release	Movement protein
Other to be classified	Host shut off	Trans-prenyltransferase
Other to be classified	Host shut off	Deoxyuridine 5'-triphosphate nucleotidohydrolase
Other to be classified	Host shut off	Ribonucleoside-diphosphate reductase
Other to be classified	Host shut off	FAD-linked sulfhydryl oxidase
Other to be classified	Host shut off	Nucleoside triphosphatase
Other to be classified	Host shut off	Alkaline nuclease
Other to be classified	Host shut off	ADP-ribose hydrolase
Other to be classified	Host shut off	dUTP nucleotidohydrolase
Other to be classified	Host shut off	Uracil-DNA glycosylase
Polyprotein	Polyprotein	Polyprotein
Protein	Proteases	Cys/Ser-proteases
Protein	Proteases	Asp-proteases
Protein	Proteases	Metallo-proteases
Protein	Proteases	Asparagine-proteases
Protein	Post translational modification	Beta-1,6-N-acetyl-glucosaminyl transferase
Protein	Post translational modification	Peptide N-myristoyl transferase
Protein	Post translational modification	thiol oxidase
Protein	Post translational modification	RNA-binding regulatory protein
Structural	Structural	Capsid protein
Structural	Structural	Nucleocapsid protein
Structural	Structural	Surface/envelope glycoprotein
Structural	Structural	Matrix protein
Structural	Structural	Tegument protein
Structural	Structural	Occlusion Body Protein
Unknown	Unknown	Hypothetical protein
Viral replication	Polymerase	DNA dependent DNA polymerase (DNA replicase)
Viral replication	Initiation	Primase
Viral replication	Initiation	Protein-primed initiation of DNA replication

Viral replication	Initiation	Protein-primed initiation of RNA replication (=VPG capping)
Viral replication	Initiation	Cress-type Endonuclease
Viral replication	Initiation	Ribozyme Endonuclease
Viral replication	Elongation	RNA-DNA duplex dependent endoribonuclease (RNase H)
Viral replication	Topology	DNA topoisomerase
Viral replication	Topology	DNA ligase
Viral replication	Topology	Helicase
Viral replication	Topology	Holliday junction resolvase
Viral replication	Topology	Integrase
Viral replication	Topology	Transposase (DDE recombinase)
Viral replication	Nucleotide metabolism	Thymidine kinase
Viral replication	Nucleotide metabolism	Thymidylate synthase
Viral replication	Nucleotide metabolism	Thymidylate kinase
Viral replication	Nucleotide metabolism	dUTP diphosphatase
Viral replication	Nucleotide metabolism	dihydrofolate reductase
Viral replication	Nucleotide metabolism	Ribonucleotide reductase
Viral replication	Regulation	E3 SUMO-protein ligase
Viral replication	Polymerase stabilization	Replication cofactor
Viral replication/transcription	Polymerase/Transcriptase	RNA dependent RNA polymerase (RNA replicase)
Viral replication/transcription	Polymerase/Transcriptase	RNA dependent DNA polymerase (Reverse transcriptase)
Viral transcription	Transcriptase	Proofreading exoribonuclease (ExoN)
Viral transcription	Viral mRNA capping	Cell-type capping
Viral transcription	Viral mRNA capping	Triphosphatase
Viral transcription	Viral mRNA capping	GTase
Viral transcription	Viral mRNA capping	N7-MTase
Viral transcription	Viral mRNA capping	2'O Mtase
Viral transcription	Viral mRNA capping	Alpha type capping
Viral transcription	Viral mRNA capping	Nido-type capping
Viral transcription	Viral mRNA capping	Mononega-type capping
Viral transcription	Viral mRNA capping	VPG capping
Viral transcription	Viral mRNA capping	Cap snatching
Viral transcription	Viral mRNA capping	Toti-type cap snatching
Viral transcription	Viral mRNA capping	FAD capping
Viral transcription	PolyAdenylation	Poly(A) polymerase
Viral transcription	PolyAdenylation	RNA polymerase stuttering
Viral transcription	Regulation	Transactivator protein
Viral transcription	Regulation	RNA export/splicing regulator
Viral-Host modulation	Anti-viral evasion	Polyuridine endonuclease
Viral-Host modulation	Anti-viral evasion	Viral ADP-Ribosylhydrolase
Viral-Host modulation	Anti-viral evasion	Asparagine deaminase

Viral-Host modulation	Anti-viral evasion	Ser/Thr kinase
Viral-Host modulation	Anti-viral evasion	Tyr kinase
Viral-Host modulation	Anti-viral evasion	Ubiquitin transferase=Ubiquitin-conjugating enzyme E2
Viral-Host modulation	Anti-viral evasion	ubiquitinyl hydrolase
Viral-Host modulation	Anti-viral evasion	Tyr/ser/thr phosphatase
Viral-Host modulation	Anti-viral evasion	2',3'-cGAMP endonuclease
Viral-Host modulation	Anti-viral evasion	3 beta-hydroxysteroid dehydrogenase/Delta 5->4-isomerase
Viral-Host modulation	Anti-viral evasion	ADP-ribose hydrolase
Viral-Host modulation	Host shut off	mRNA-decapping
Viral-Host modulation	Apoptosis modulator	vCyclin
Viral-Host modulation	Apoptosis modulator	Apoptosis/cell-cycle modulator
Viral-Host modulation	Apoptosis modulator	vBcl-2
Viral-Host modulation	Anti-viral evasion	Interferon antagonist
Viral-Host modulation	Anti-viral evasion	Cytokine modulator
Virion assembly	Assembly	Assembly factor
